

Inequality and Wage Dispersion

Labor I — Week 8

Teaching deck draft

Central question and why this week is heavier

- Week 8 asks how labor economists should measure inequality and organize the mechanisms behind wage and earnings dispersion.
- This week is heavier because it consolidates Weeks 1–7 and sets up Weeks 9–10.
- We need to separate measurement objects, descriptive decompositions, causal mechanisms, and welfare interpretation.

Bridge from Weeks 5–7

- Week 5: wages reflect skill, sorting, and sometimes rents.
- Week 6: households transform wage opportunities into hours, earnings, and participation.
- Week 7: jobs are wage-plus-amenity bundles, so wage inequality is not automatically welfare inequality.

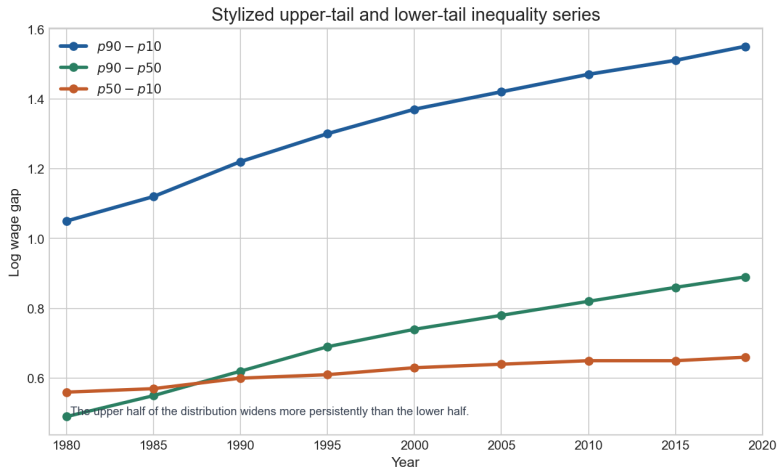
Inequality objects

Object	What it captures	What it misses
Hourly wages	Wage-setting and skill prices	Hours, employment, amenities
Annual earnings	Wages plus work quantity	Nonwage job quality
Hours / employment	Quantity margins	Wage-setting itself
Total job value	Wages plus amenities and stability	Hard to observe directly
Household income	Pooling and transfers	Individual labor-market mechanisms

$$q_t(\tau) = F_t^{-1}(\tau), \quad D_t^{90-10} = q_t(0.9) - q_t(0.1)$$

- Quantile gaps let us separate upper-tail and lower-tail inequality.
- $p_{90} - p_{50}$ and $p_{50} - p_{10}$ answer different questions.
- Variance objects and top shares emphasize different parts of the distribution again.

Percentile-gap series



Between-group versus within-group inequality

$$\text{Var}(\log w) = \text{Var}(\mathbb{E}[\log w \mid g]) + \mathbb{E}[\text{Var}(\log w \mid g)]$$

- Inequality can rise because observed group means diverge.
- It can also rise because dispersion within groups increases.
- This is descriptive accounting, but it strongly disciplines mechanism choice.

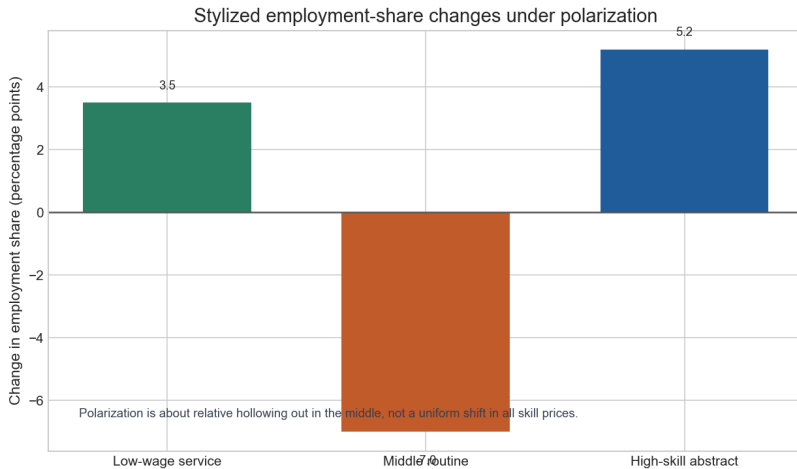
$$\log w_{it} = X'_{it}\beta_t + u_{it}$$

- Observable characteristics explain only part of dispersion.
- The residual is not one structural force; it mixes unobserved skill, rents, institutions, and measurement limits.
- Juhn-Murphy-Pierce made this residual component central to the inequality conversation.

Canonical distributional facts

- Upper-tail inequality rises strongly after 1980.
- Lower-tail changes are more episodic and institution-sensitive.
- The college premium rises, but it is not the entire story.
- Composition-adjusted and raw changes need to be distinguished.

Tasks and polarization



Supply, demand, and skill-premium logic

- Relative demand shifts can raise the return to skill when supply does not keep up.
- This logic connects Week 4 human capital directly to Week 8 inequality.
- It is strongest for broad skill premiums, but weaker as a complete explanation for every tail of the distribution.

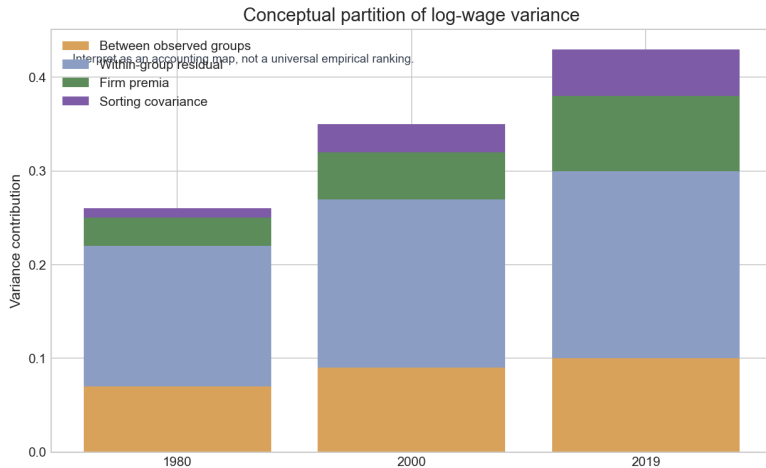
Institutions and the lower tail

- Minimum wages, unions, and wage-setting rules often matter most for the lower tail and middle.
- Institutional compression changes some percentiles more than others.
- Counterfactual distribution work helps separate composition from institutional environments.

$$\log w_{it} = \alpha_i + \psi_{J(i,t)} + X'_{it}\beta + \varepsilon_{it}$$

- Modern linked employer-employee data adds the employer as an explicit source of wage dispersion.
- Inequality can reflect worker heterogeneity, firm premia, and positive sorting between them.
- Firm inequality is one way of understanding worker inequality, not a separate topic.

Conceptual inequality decomposition



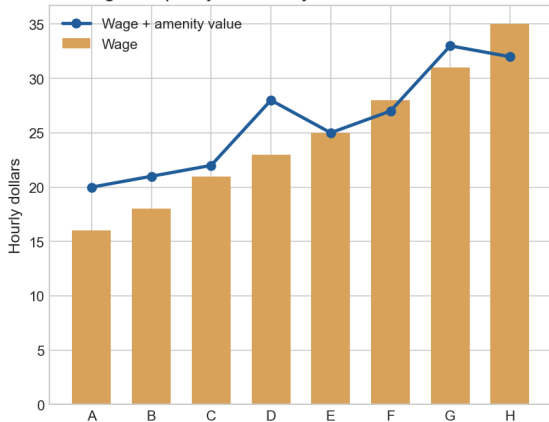
Wages versus total labor-market value

$$v_i = w_i + a_i$$

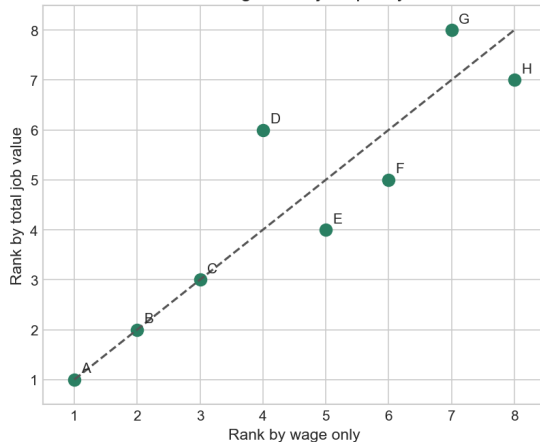
- a_i captures amenities, safety, schedule quality, stability, or other nonwage job value.
- Wage inequality can understate or overstate inequality in total labor-market value.
- This is the Week 7 to Week 8 bridge.

Wage ranking versus total job value

Wage inequality and total job value need not match



Ranks can change once job quality is valued



Empirical toolkit map

Method	Main use	Data need	Main limit
Percentile series	Track distributional change	Repeated wage data	Descriptive only
JMP residual decomposition	Separate observables and residuals	Rich micro covariates	Residual mixes mechanisms
DFL reweighting	Counterfactual distributions	Cross-sections with overlap	Still partly descriptive
RIF methods	Unconditional quantile effects	Micro data plus IF methods	Interpretation depends on functional
AKM decomposition	Worker, firm, sorting terms	Linked worker-firm panel	Mobility and design sensitivity

- Group gaps in Week 9 sit inside this wider distributional architecture.
- Discrimination is not the same thing as general dispersion in wages, firms, tasks, or places.
- Segmentation can operate through amenities and access to high-premium firms as well as through posted pay.

- Inequality is multi-object: wages, earnings, employment, and total job value can diverge.
- Quantile gaps, variance decompositions, and firm decompositions answer different questions.
- No single mechanism explains the post-1980 distribution.
- Week 8 is the architecture week for reading discrimination, mobility, and frontier inequality work carefully.

Appendix: optional extension block

- Linked employer-employee data and AKM implementation details.
- Structural interpretations of firm premia, rents, and imperfect competition.
- Supply, institutions, and firms across the full wage distribution.